

Networking Fundamentals Notes

I. Network Basics

Key Components of a Home Network

- **ISP (Internet Service Provider)**: The company you pay for internet access
- **Router**: Connects all devices to the internet (wireless or Ethernet)
- **WAN (Wide Area Network)**: Network between router and broader internet
- **WLAN (Wireless Local Area Network)**: Network between router and wireless devices
- **LAN (Local Area Network)**: Network between router and wired devices
- **Hosts**: Each machine on a network

Data Transmission

- Data travels through networks as **packets**
 - Packets contain the information being transmitted between hosts
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2. OSI Model

Overview

- **OSI (Open Systems Interconnection)**: Theoretical networking model
 - Shows packet traversal through **7 layers**
 - Foundation for the TCP/IP model used today
 - More theoretical than practical for modern networking
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3. TCP/IP Model

Four-Layer Structure

The TCP/IP model is the actual implementation of networking that the Internet uses.

Application Layer (Top Layer)

Purpose: Interface between computer programs and transport layer

Protocols:

- **HTTP (Hypertext Transfer Protocol):** Used for webpages
- **SMTP (Simple Mail Transfer Protocol):** Used for email transmission

Transport Layer

Purpose: Determines how data is transmitted, ensures data integrity

Protocols:

- **TCP (Transmission Control Protocol):** Reliable data delivery
- **UDP (User Datagram Protocol):** Unreliable but faster data delivery

Network Layer

Purpose: Moves packets between hosts and across networks

Protocols:

- **IP (Internet Protocol):** Routes packets between machines
- **ICMP (Internet Control Message Protocol):** Error messages and debugging

Link Layer (Bottom Layer)

Purpose: Sends data across physical hardware

Examples: Ethernet, fiber optic cables

4. Network Addressing

Three Types of Network Identifiers

MAC Addresses

- **Purpose:** Unique hardware identifier that never changes
- **Format:** 00:C4:B5:45:B2:43
- **Example:** Dell devices have OUI 00-14-22, so a Dell adapter might be 00-14-22-34-B2-C2
- **Location:** Network interface card (NIC)

IP Addresses

- **Purpose:** Software-based device identification on networks

- **Format:** 10.24.12.4 (IPv4 example)
- **Characteristics:** Can change, unique across the Internet
- **Example:** 192.12.41.4

Hostnames

- **Purpose:** Human-readable names for IP addresses
 - **Example:** Instead of remembering 192.12.41.4, use myhost.com
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5. Application Layer Deep Dive

Email Example: Sending to Patty

Process:

1. Email client uses SMTP protocol
2. Data sent through **port 25** (SMTP port)
3. Application layer encapsulates data
4. Data passed to Transport layer for segmentation

Packet Structure

- **Header:** Contains routing information (where packet goes/comes from)
 - **Payload:** Actual data being transferred
 - **Terminology:** Called "packet" generally, "segment" in transport layer, "frame" in link layer
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6. Transport Layer Deep Dive

Segmentation

- Breaks data into **segments** for easier transport
- Segments reassembled in correct order at destination

Ports

- **Purpose:** Direct data to specific processes/services
- **Example:** HTTP uses port 80, SMTP uses port 25
- Transport layer attaches source and destination ports

UDP vs TCP

UDP (User Datagram Protocol)

- **Characteristics:** Unreliable, faster
- **Use Case:** Media streaming (okay to lose some frames for speed)

TCP (Transmission Control Protocol)

- **Characteristics:** Reliable, connection-oriented
- **TCP Handshake Process:**
 1. Client sends **SYN** segment to server
 2. Server sends **SYN-ACK** to acknowledge
 3. Client sends **ACK** to complete connection
- **Features:** Uses sequence numbers to ensure correct order

Email Example Continued

Transport layer attaches destination port (25) and source port for the email data.

7. Network Layer Deep Dive

Routing Function

- Determines packet routing from source to destination
- Internet consists of many interconnected **subnets**
- **Example:** Enables access to <www.google.com> even though it's on a different network

IP Packet Creation

- Encapsulates transport layer segment in IP packet
 - Attaches source and destination IP addresses
 - Packet now knows where it's going and where it came from
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8. Link Layer Deep Dive

Frame Creation

- Encapsulates packet into a **frame**
- Adds source and destination MAC addresses
- Includes checksums and packet separators

ARP (Address Resolution Protocol)

Purpose: Finds MAC address associated with an IP address

Process:

1. Check ARP lookup table first
2. If not found, send broadcast message to network
3. Host with requested IP responds with MAC address

Example: "Which host has IP 10.10.1.4?" - broadcast to all network hosts

9. Complete Packet Traversal Example

Pete Sends Email to Patty

Outbound Journey (Pete's side)

1. **Application:** Email data sent to transport layer
2. **Transport:** Data encapsulated in TCP segment with ports
3. **Network:** Segment encapsulated in IP packet with IP addresses
4. **Link:** Packet encapsulated in frame with MAC addresses

Inbound Journey (Patty's side)

1. **Link:** Frame received, integrity checked, de-encapsulated
 2. **Network:** IP packet examined, destination IP verified, de-encapsulated
 3. **Transport:** Segment de-encapsulated, port numbers checked
 4. **Application:** Final email message presented to Patty
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10. DHCP (Dynamic Host Configuration Protocol)

Purpose

Automatically assigns network configuration to devices:

- IP addresses
- Subnet masks
- Gateways

Analogy

Like a phone carrier assigning phone numbers - you get a "lease" on an IP address for a specific time period.

DHCP Process (4-Step Handshake)

1. **DHCP DISCOVER:** Client broadcasts search for DHCP server
2. **DHCP OFFER:** Server responds with IP address, subnet mask, lease time
3. **DHCP REQUEST:** Client broadcasts acceptance of specific offer
4. **DHCP ACK:** Server sends acknowledgment

Typical Setup

- **Home networks:** Router usually acts as DHCP server
 - **Benefits:** Prevents duplicate IP addresses, reduces admin workload
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Key Takeaways

1. **Layered Approach:** Each layer adds specific information to packets
2. **Dual Addressing:** Both MAC (hardware) and IP (software) addresses needed
3. **Packet Journey:** Data travels through all layers twice (send and receive)
4. **Automatic Configuration:** DHCP handles network setup automatically
5. **Protocol Cooperation:** Multiple protocols work together at each layer